

SUPPORT BOT FOR AUTOMATING L1 TRIAGE

A Project Report Submitted in Partial Fulfillment of the Requirement of the Degree

Of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

HARSH AGARWAL (201600622)

GROUP: 16

Under the guidance of

Sailaja Karri ,

Senior Manager, DevOps Onboarding,

DAO Dell, Digital and IT

Arun Kumar Singh

Assistant Professor,

E&C Dept, SMIT



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MAJITAR, EAST SIKKIM-737136, June 2020

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MAJITAR, EAST SIKKIM-737136, June 2020

CERTIFICATE

This is to certify that the project report entitled “***Support Bot for Automating L1 Triage***” submitted by Harsh Agarwal (201600622) to Sikkim Manipal Institute Of Technology, Sikkim in partial fulfillment for the award of degree of Bachelor of Technology in Electronics And Communication Engineering, is a bonafide record of the project work carried out by him under my guidance and supervision during the academic session January– May, 2020.

Arun Kumar Singh

Assistant Professor

Dept. Of Electronics & Communication

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H.O.D

Dept. Of Electronics & Communication

Sikkim Manipal Institute of Technology

Dear Harsh Agarwal,

This is to confirm that you have been interning with Dell Technologies.

Your Internship details are mention below :

Project Name	Support BOT for Automating L1 Triage
Project Start Date	January 20, 2020
Project End Date	May 29, 2020
Location	Hyderabad
Reporting Manager	Karri Sailaja
Guide Name	Subramanyam Kirikera
Sponsor Name	Cody Taylor

You have successfully completed the project work with Dell Technologies. Your findings in course of the project has been found to be practical and relevant. Some of the recommendations will be incorporated on approval from the business.

Your performance during the tenure of the internship has been found to be satisfactory.

Note: You can use this email as a confirmation/certificate of internship completion in case the college requires.

Best Regards,
Dell University Relations – India
[Dell Technologies](#) | Talent Acquisition
For any queries reach out - UR_India@Dell.com

ABSTRACT

The report provided herewith is an attempt to give a detailed description of the objective, approach and the solution to the problem dealt with through the project entitled “Support Bot for Automating L1 Triage”. This project has been developed during the internship period at DELL Technologies Ltd. and as a Final Year Project.

Dell as a company is investing heavily in its digital arm and Dell Digital is undergoing a significant transformation towards innovating more effective and profitable ways of product development. Development Operations is at the core of this transformation and is essential in a way that very few other teams are because every single application has to go through this channel before deployment of any kind. This project is a stepping stone towards making this process efficient and cost-effective. It is aimed at automating the task of responding to Support and Onboarding requests raised for the DevOps team, on the Service Now portal.

The tickets concerned here are raised at an unprecedented pace and usually number in the thousands every month. The manual handling of these tickets is a tedious process and does not do justice to the technical acumen of a capable engineer. It is also revenue degrading from a company's point of view. Hence, it becomes necessary to find a solution. The problem is dealt with, using automation in the form of a ChatBot and is developed on the Visual Studio IDE. The programming language used is primarily C#.

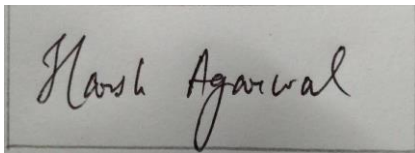
Acknowledgement

I would like to express my sincere gratitude and appreciation to the Electronics and Communication Department, SMIT, for providing me with the opportunity of undertaking a major project at Dell Technologies and to our HOD Dr. Saurab Dhar, for his support.

I would like to mention my special thanks to my project guide, Arun Kumar Singh, for his meaningful contribution to this work.

I would also like to thank my manager Mrs. Sailaja Karri and Mr. Subrahmanyam Kirikera who acted as my mentor throughout the internship and was a constant support.

Finally, I would like to extend my gratitude to all my batchmates and friends at Sikkim Manipal Institute of Technology, for providing their support in various manners during my internship.

A rectangular box containing a handwritten signature in cursive script that reads "Harsh Agarwal".

HARSH AGARWAL

(201600622)

Dept. Of Electronics and Communication, SMIT.

List of Abbreviations

DEVOPS - Development and Operations

SNOW- Service Now

API- Application Programming Interface

IDE- Integrated Development Environment

TFS- Team Foundation Server

POC- Proof of Concept

CI - Continuous Integration

CD - Continuous Deployment

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Chapter 1 - Introduction

Development Operations is an important link in the chain of processes which a software needs to go through before being deployed or be commercially released. One of the main functions of the DevOps team is to respond to the requests raised or the incidents reported by the other teams involved in the development or testing or deployment of the software or a piece of code. These requests or concerns are raised as tickets through a portal named Service Now. The current practice to handle these tickets involves a person (L1 level). The aim of the project is to remove this requirement by automating the process.

It is imperative to understand the need and the usage of the application under consideration. This project was initiated in order to aid the support system provided by the DevOps team, while reducing the workforce invested in it and improving the efficiency. This application takes in the tickets and sorts it into the major fields which we will explore in the later part of the report and then, the issue is resolved by keeping the manual involvement to the minimum. This application acts as a one stop solution for the resolution of the issues based on the type of issue ticket raised by the end user.

1.1 Introduction to DevOps

Typically, DevOps is not a tool or software; it is a culture that we can adapt to foster ongoing improvement. DevOps forms a channel to communicate and integrate between the developer and the operations team to enhance collaboration and productivity.

In DevOps, the development and testing phase of an application run parallelly or go hand in hand. The developers and testers work as a team on the deliverables with direct communication by identifying and resolving issues. They also take feedback which is an ongoing process and one which significantly impacts the results.

1.2 Objective

The overall objective of the project can be summarized as:

“Minimization of DevOps Support effort by streamlining of the process using automation of the possible steps and the introduction of a BOT for applicable support requests.”

With this application we aim to achieve a service which will run on some server and be always available to cater the needs of the people raising queries in forms of tickets for assistance regarding **Access Request, Onboarding** and **Support Request**. The project will serve the following dimensions:

- To quickly fetch and resolve the tickets.
- To segregate the tickets based on priorities and kind of assistance required
- To ensure minimal manual intervention in the resolution of the ticket.
- Diverting the issue to a support person if the problem involved is beyond the bot`s ambit.

1.3 Overview

The few critical aspects of this project are to create an automated backend in order to cater to the incoming tickets and intelligently mapping them to their respected resolving branch, the creation of a communication medium between the involved parties and resolution of the ticket, if possible. Its key features include:

- Real time fetching of the tickets
- Access to Service Now(SNOW), Team foundation Server (TFS), Chat Applications.
- Initiation of conversation channels
- Keeping the support team and the end user, who is raising the tickets, in a closed knit, well-informed environment

This project demands an application which won't be a separate portal for the people who want to raise a ticket Rather it would act as a backend patch which removes the intervention of manual labor for the resolution of the issue. Hence this application won't be directly available to the user base rather will be triggered every time in the backend and only a small part of it will be interacting with the user base in the form of a chat bot.

1.4 What Is A Chatbot?

At the most basic level, a chatbot is a computer program that simulates and processes human conversation (either written or spoken), allowing humans to interact with digital devices as if they were communicating with a real person. Chatbots can be as simple as rudimentary programs that answer a simple query with a single-line response, or as sophisticated as digital assistants that learn and evolve to deliver increasing levels of personalization as they gather and process information.

1.4.1 How do ChatBots Work?

Driven by AI, automated rules, natural-language processing (NLP), and machine learning (ML), chatbots process data to deliver responses to requests of all kinds.

There are two main types of chatbots.

- **Task-oriented (declarative) chatbots** are single-purpose programs that focus on performing one function. Using rules, NLP, and very little ML, they generate automated but conversational responses to user inquiries. Interactions with these chatbots are highly specific and structured and are most applicable to support and service functions—think robust, interactive FAQs. Task-oriented chatbots can handle common questions, such as queries about hours of business or simple transactions that don't involve a variety of variables. Though they do use NLP so end users can experience them in a conversational way, their capabilities are fairly basic. These are currently the most commonly used chatbots.
- **Data-driven and predictive (conversational) chatbots** are often referred to as virtual assistants or digital assistants, and they are much more sophisticated, interactive, and

personalized than task-oriented chatbots. These chatbots are contextually aware and leverage natural-language understanding (NLU), NLP, and ML to learn as they go. They apply predictive intelligence and analytics to enable personalization based on user profiles and past user behavior. Digital assistants can learn a user's preferences over time, provide recommendations, and even anticipate needs. In addition to monitoring data and intent, they can initiate conversations. Apple's Siri and Amazon's Alexa are examples of consumer-oriented, data-driven, predictive chatbots.

Advanced digital assistants are also able to connect several single-purpose chatbots under one umbrella, pull disparate information from each of them, and then combine this information to perform a task while still maintaining context—so the chatbot doesn't become “confused.”

1.4.2 The Value Chatbots Bring to Businesses and Customers

Chatbots boost operational efficiency and bring cost savings to businesses while offering convenience and added services for customers. They allow companies to easily resolve many types of customer queries and issues while reducing the need for human interaction.

With chatbots, a business can scale, personalize, and be proactive all at the same time—which is an important differentiator. For example, when relying solely on human power, a business can serve a limited number of people at one time. To be cost-effective, human-powered businesses are forced to focus on standardized models and are limited in their proactive and personalized outreach capabilities.

By contrast, chatbots allow businesses to engage with an unlimited number of customers in a personal way and can be scaled up or down according to demand and business needs. By using chatbots, a business can provide humanlike, personalized, proactive service to millions of people at the same time.

Consumer research is showing that messaging apps are increasingly becoming the preferred method for connecting with businesses for certain types of transactions. Delivered through messaging platforms, chatbots enable a level of service and convenience that in many cases exceeds what humans can provide. For example, banking chatbots save an average of four minutes per inquiry compared to traditional call centers. The same capabilities that help businesses achieve greater efficiency and cost reductions also deliver benefits to customers in the form of an improved customer experience.

1.5 Dell Service Now Page

This is the IT service page accessible to all the teams of Dell International Pvt. Ltd. This is the entry node for any request that a person wants to raise. The customer may select the type of assistance he/she wants and then follow up with the general flow and finally submit his/her query to raise a ticket. This is a very simple service portal and easy to navigate through.

DevOps Support For Gitlab, Octopus, Teamcity, Artifactory and other CICD Tools [AUTO]

Access and support for various DevOps tools

Tools:

CICD tools

1. GitLab : <http://gitlab.dell.com>
2. TeamCity : <https://dctestteamcity.dell.com>
3. Octopus Deploy : <https://dcsoctopus.dell.com>
4. Artifactory : <https://dcsartifacts.dell.com>
5. SonarQube : <https://sonarqube.dell.com>
6. Selenium Grid : <https://selenium-grid.ausmpc01.pcf.dell.com>
7. HashiCorp Vault : <https://vault.dell.com/>
8. SauceLabs : <https://tunnelmanager.ausmpc01.pcf.dell.com>

Online Documentation

1. Confluence : <https://confluence.dell.com>

Online support documents

1. GitLab : <https://confluence.dell.com/pages/viewpage.action?pagelId=174883842>
2. SonarQube : <https://confluence.dell.com/display/DD/SonarQube+Scan>
3. Selenium Grid : <https://confluence.dell.com/display/DD/DevOps+Selenium+Grid>
4. SauceLabs : <https://confluence.dell.com/display/DD/SauceLabs>
5. HashCorp Vault : <https://confluence.dell.com/display/DD/DAO+HashiCorp+Vault>

Fig.1.1: Tools under DevOps team

Requested For

* Who is this for?

Abhishek Nayak

×

▼

* Type of Service

Access Request

▼

* Tool Name ?

-- None --

▲

Q

-- None --

▲

Artifactory

Confluence

GitLab

HashiCorp Vault

Octopus

SauceLabs

Fig.1.2: Ticket generation for Access Request

Requested For

* Who is this for?

Abhishek Nayak ✕ ▼

* Type of Service

Onboarding CI/CD Tools ▼

* Tool Name ?

-- None -- ▲

— None —

GitLab

HashiCorp Vault

SauceLabs

Fig.1.3: Ticket generation for Onboarding Request

Requested For

* Who is this for?

Abhishek Nayak ✕ ▼

* Type of Service

Support Request ▼

* Tool Name ?

-- None -- ▼

* Application Name

* Detailed Request Description

* Severity

-- None -- ▼

Fig.1.4: Ticket generation for Support Request

Chapter 2 – Motivation

The motivation for this project is saving human work hours. In the present scenario, a technical supervisor is constantly deployed to check and handle tickets which are usually raised at an unprecedented pace on the Service Now (SNOW) portal. This task is not only tedious and monotonous but also involves a poor utilization of the technical skills of an individual. By automating the process, an individual's technical acumen can be utilized in a more meaningful manner and the ticket handling process can be speeded up which in turn will speed up the entire software deployment or software production chain.

2.1 Bridging the Gap – Importance of DevOps

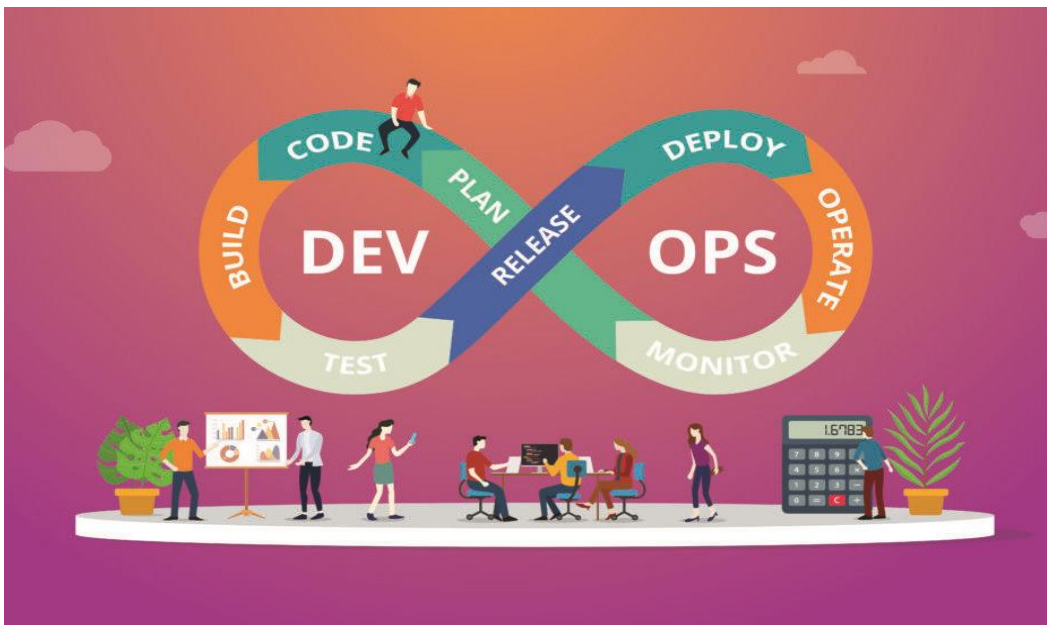


Fig 2.1 : DevOps

Typically, DevOps is not a tool or software; it is a culture that we can adapt to foster ongoing improvement. DevOps forms a channel to communicate and integrate between the developer and the operations team to enhance collaboration and productivity.

DevOps is an approach and set of processes which brings the development and operations team together to achieve the assigned task. It fills the gaps which are found in the traditional software development approach (waterfall method) thus leaving more scope for improved efficiency, updating and delivering the application at a much faster pace. Hence, DevOps is gaining more popularity.

Some important features of **DevOps** -

- Quick on-time delivery with no compromise on quality.
- Gives more scope of testing at the end of each phase.
- Cross-functional skills, to learn improved practices.
- Fosters better relation between development and operations team.
- 60% less time to market.
- Shorter time to recover and fix bottlenecks.

2.2 Why Automation?

Automation is a broad term that can cover many areas of technology where human input is minimized. This can include everything from business-specific types such as: business process

automation (BPA), IT automation, marketing automation and industrial automation. It also covers personal applications such as home automation.

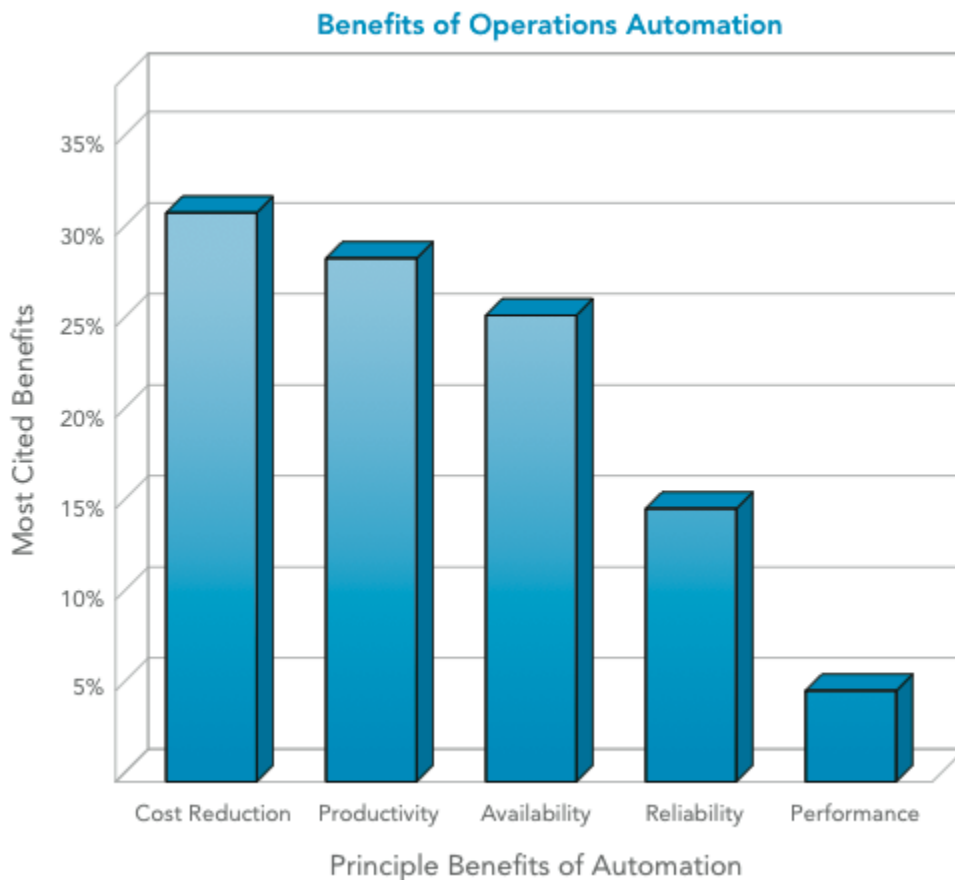


Fig 2.2 : Principal Benefits of Automation

2.2.1. Reducing Operational Costs

Every business faces global pressure to increase their profitability. One approach is to reduce costs. But, reducing the capabilities of the computer center negatively impacts the entire company.

Automation software is a better and more intelligent approach to cost containment and reduction. The greatest opportunity is to increase service to the customer (end user) while systematically reducing costs. Management often overlooks this potential for savings. Most modern servers have a low operating cost and the total cost of ownership has been declining. Even so, the cost of the operations staff can be as high as 71% of the total cost.

2.2.2 Increasing Productivity

As an organization's technology demands grow, productivity becomes a bigger concern. Typically, as other business areas were given tools to increase their productivity and effectiveness, IT operations took a back seat. The proliferation of desktop productivity software has created substantial gains in the office and HR environments. But, instead of alleviating workload for the IT professionals in the back room, the spread of PCs has meant more tasks to be accomplished.

As people use computers more, they place greater demands on the system. More users are generating more jobs, and printed output has increased despite efforts to reduce printed reports. In spite of the trend to online transaction-oriented and client/server systems, batch workloads continue to grow. Production batch jobs still consume the majority of CPU time, and in large shops, jobs are constantly being added.

Automated operations can resolve these issues in several ways.

Job scheduling software increases batch throughput by automating the production batch schedule. In the early days, computer throughput was limited by how fast operators could reset switches on the console. The solution for today is not to allow the computer to remain idle while

waiting for the operator to release the next job. You save time and money by eliminating the lag time between jobs and minimizing operator intervention. In some instances, you can trim hours off your process, which lets you process more work and significantly improve system use.

Once the job schedule is established, the automation software executes the commands precisely and in the correct sequence, eliminating operator errors. Forecasting job completion and being able to perform “what if” analyses of schedule changes benefits operations by removing much of the guesswork from daily tasks.

2.2.3 Ensuring High Availability

Companies are continually more reliant on their computers. Day-to-day business is routinely conducted with online systems: order entry, reservations, assembly instructions, shipping orders—the list goes on. If the computer is not available, the business suffers.

Years ago, it was considered acceptable to have the computer unavailable for a few hours.

Today, with the high volume of cloud computing, the outage of key systems can cost millions of dollars in lost revenue and tarnish a company’s reputation.

High availability is clearly one of IT management’s primary goals. Here too, automated operations can help. A disk drive may crash, but the situation becomes serious when there is not an adequate backup— or worse, the tape cannot be found. A key advantage to automation is the ability to automate your save and recovery systems to ensure protection from the potential disaster of disk loss, or inadvertent damage to system objects from human error.

In a networked environment, centralized management also makes sense. Remote resources can solve business issues while a single operator at a central console observes critical functions throughout the network. Continuous monitoring with a low CPU and communications overhead makes it easier to spot vital network performance trends.

2.2.4 Increasing Reliability

Productivity is an obvious benefit of automation. However, reliability is the real gem that sparkles with automation. It is the cornerstone of any good IT operations department and without it you have confusion, chaos, and unhappy users. IT operations requires two opposed skill sets: On one hand, an operations person needs highly technical skills, such as the ability to understand the complexities of an operating system and to analyze and solve problems as they arise. On the other hand, this same person has to be content pushing buttons and loading paper.

Let's face it, off-shift operations include some of the most boring, repetitive, and error-prone tasks of an IT organization. But, when you remove the human factor, you eliminate most batch processing errors.

Automated operations ensure that jobs are not forgotten or run out of sequence, that prerequisite jobs are completed successfully, that the input data is correct, and that any special processing is performed.

All these types of errors occur in single-location organizations. Now, imagine a network of multiple systems, geographically dispersed that include multiple operating systems, communications issues, integrated local area network processing, and attached PCs. The chance

for errors rises exponentially. The only way to make this type of environment work is automated operations.

Software can handle complex tasks dynamically and intelligently, based on predefined parameters. Yet, critical company functions such as releasing jobs, performing backups, and ensuring communications, are normally performed by entry-level individuals within the IT organization. The benefit of an automated system is that these functions are reliably executed by the automation software, relieving operations personnel from hours of tedious, boring and manual tasks.

2.2.5 Optimizing Performance

Every company would like to have their enterprise perform like a thoroughbred. In reality, it is more likely to be overburdened with work. Even though advancements in computers make them faster and less expensive every year, the demands on them always catch up and eventually exceed the level of capability that a company's computer infrastructure possesses. That leaves a lot of companies wanting to improve their system performance.

Two options to improve performance are to upgrade hardware or purchase a newer system—both expensive choices. It's also possible to tune a system for better performance, but this takes a highly skilled person who is not normally available 24 hours a day. And, once a system is tuned for a specific workload, if the workload changes, the settings are no longer optimum

2.3 Summary

The DevOps team support group includes people who are responsible for the resolution of these requests raised in the form of tickets and each person in that group spends quite a lot of time in doing something which is not technically taxing and can be easily done, like granting access or resolving issues which are just a button's click away. The inefficient use of human resources and the amount of time which could have been used to accomplish something else, planted the idea of automating these tasks via a backend bot. So, the idea of developing a bot was discussed and greatly appreciated. This would not only save a lot of time and workforce but would also simplify the process.

Chapter 3 - Literature Review

SL. NO.	Author	Title	Year	Findings	Relevance to the project
1.	ZHAOSHENG ZHU ; GUOHAN LU ; YAN CHEN ; ZHI JUDY FU ; PHIL ROBERTS ; KEESOOK HAN, , <i>2008, 32nd Annual IEEE International Computer Software and Applications Conference: 425-436</i>	Botnet Research Survey	2008	The paper presents a survey of recent advances in botnet research	Helpful in developing a general idea of internet bots.
2.	Andreas Schaefer, Marc Reichenbach Dietmar Fey, 2012, In: Proceedings of IEEE Visualization, pp 190–198 .	Continuous Integration and Automation for Devops	2016	This work presents insights in how automation can be used to improve scalability and testability while simultaneously reducing the operators’ work.	Provides a general idea of Development Operations and its need in the software development cycle.
3.	WU YU-PENG,TAO JUN- CAI (COMPUTER CENTER, NANCHANGUNIVERSITY), <i>In: Proceedings of the 21st conference on Large Installation System Administration Conference, Berkeley, CA, USA, USENIX Association, pp 21:11:13</i>	Research and Development of Data Exchange on Automatic Programming Systems.	2020	A new solution based on XML is presented in order to improve automatic programming system.	Utilisation of dot net framework to develop internet bots.

Table 1: Literature Review

Chapter 4 - Problem Definition

Tickets are raised on the SNOW portal which are checked by software engineers from time to time. The tickets can be of various types- access request, onboarding request or service request. The person who handles the tickets is referred to as L1 engineer. The L1 engineer segregates the ticket and assigns it to a L2 engineer who is adept at solving the problem and if still unresolved, the problem is passed onto a L3 engineer who is specialized in solving the issue raised in the particular ticket. The problem is to automate the L1 engineer's job and, if possible, resolve the issue through the bot without having to pass it onto the L2 level.

4.1 Blueprint and Idea

In the following flow diagram, we can see that there is a customer who will raise the ticket(the customer is generally a Dell employee belonging to some other team and who wants assistance from the DevOps team) with an issue either related to some of the tools which are hosted on the server or needing assistance with hosting of a new application onto the server.

So, the customer goes to the *Service Now* page and raises the ticket based on whether the issue is related to *Access Request*, *Onboarding Request* or *Support Request*.



Fig. 4.1: Flowchart showing the flow of tasks to be carried out

Primarily, the customer goes onto the Service Now page and selects one of the above three broad categories of request and then, after selecting one of them, he/she is directed to a questionnaire which asks questions regarding the mandatory information required for the request being raised.

As soon as the user/customer raises a ticket based on one of the three branches, a bot will be initiated, which will cater to the needs of the customer. The bot will invite that customer to a channel on Slack using a channel invite link mailed directly to the customer's outlook mailbox. This promotes security as no one else can access the chat room. In the chat room, the bot can ask specific questions related to the type of assistance requested by the customer. This is a good approach as the conversation is recorded and kept till the channel expires.

If the bot resolves the issue and the customer is satisfied then the channel will be closed and the ticket will be closed, but if the request requires some additional assistance in the form of human intervention, a support executive is added to the same channel via a link sent to the executive via outlook mailbox and then he can take over from there.

4.2 Basic Initiation

When a customer goes onto the Service Now page and initiates a ticket, it is fetched by a *service function* which is an extension to the Support Bot and whose function is to constantly fetch tickets generated on the page.

These tickets are kept in a *ticket rendering queue* which will ensure that the tickets are fetched in a *first-in-first-out* manner. These tickets are then passed through a sorter mechanism which will sort the tickets into the three different branches depending on the header of the tickets.

Technology Involved –The bot extends this backend service function as a form of an *API* which will form a bridge and provide access to the Service Now page. This *API* serving mechanism is then clubbed with an iterative loop with a self-calling mechanism which will ensure continuous fetching of the tickets. Then the sorting mechanism will read the ticket header, which contains the type of request it belongs to, and will sort the ticket into the respective branches.

4.3 Access Request

This request helps in gaining access to some of the DevOps tools which are already hosted on the server like Confluence, Gitlab, Artifactory. The owner or maintainer has not made the tool open and free to use and hence the user has to go through the Service Now page in order to raise a ticket requesting access to use that tool.

The basic procedure is that the customer goes on to the Service Now page and choose the Access Request ticket from the list, post that he/she is provided with a list of tools that are available on the server and from that list the user selects the tool that he/she has demand for. Post choosing the tool a ticket is generated which is then assigned to a support executive to go through the request and take necessary steps in order to grant the access.

Technology Involved: After the sorting function, the tickets are stored in a special *Access request* only queue. The bot framework selects tickets one by one from the queue and then depending on the request for the tool the framework navigates to that specific tool and grants access by clicking the access button. This way the access to the tool is provided to the customer via bot and without any human intervention.

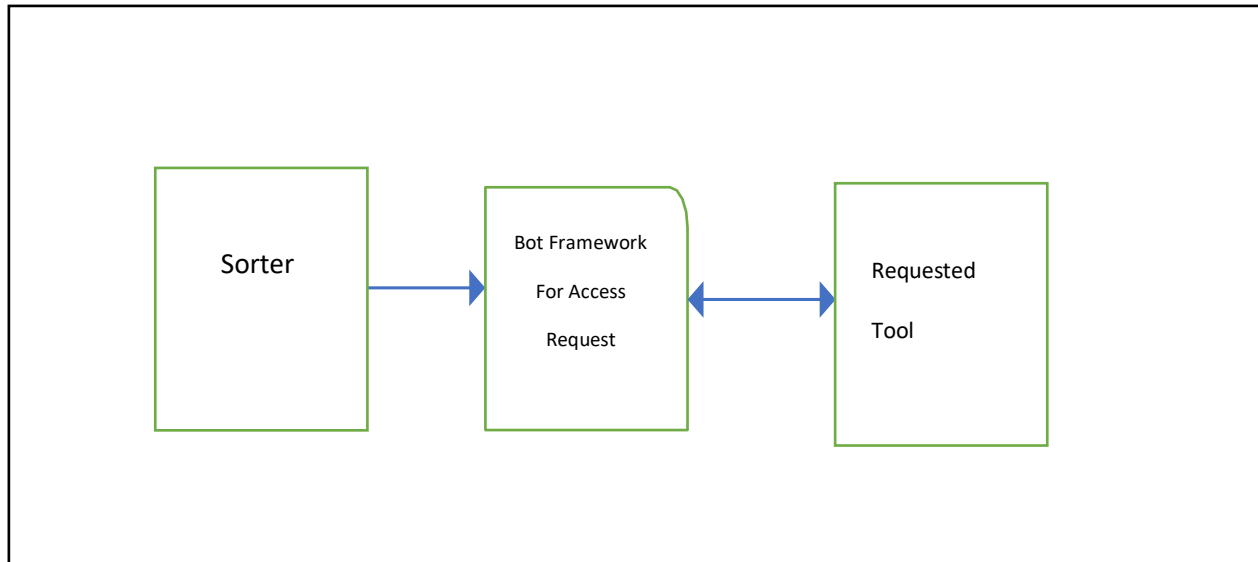


Fig.4.2: Access Request flow

4.4 Support Request

Here, the customer raises a ticket to gain support regarding some of the DevOps related tools.

Technology Involved: After the ticket is raised and sorted, the *Support bot* catches the request and then creates a Slack channel for the conversation with the customer. The customer is sent a link to the channel via outlook mailbox so as to keep it secured and then the customer joins the channel. The channel consists of the bot and the customer who has raised the ticket. Now the bot will be posting questions to the channel. After the customer has answered the questions on the channel, the bot is triggered again in order to intimate the L1 support executive.

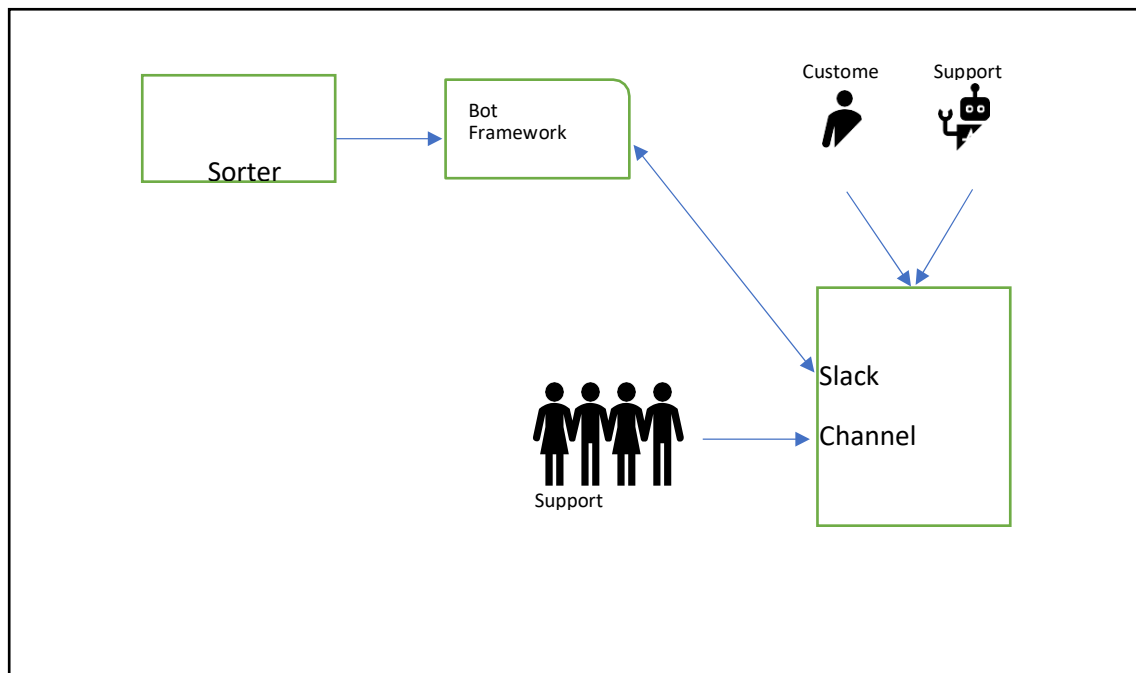


Fig 4.3: Support Request Flow

4.5 Onboarding Request

Onboarding process/request is raised when a customer wants to deploy some application or tools onto Gitlab and wants assistance with the Continuous Integration & Continuous Development (CI/CD).

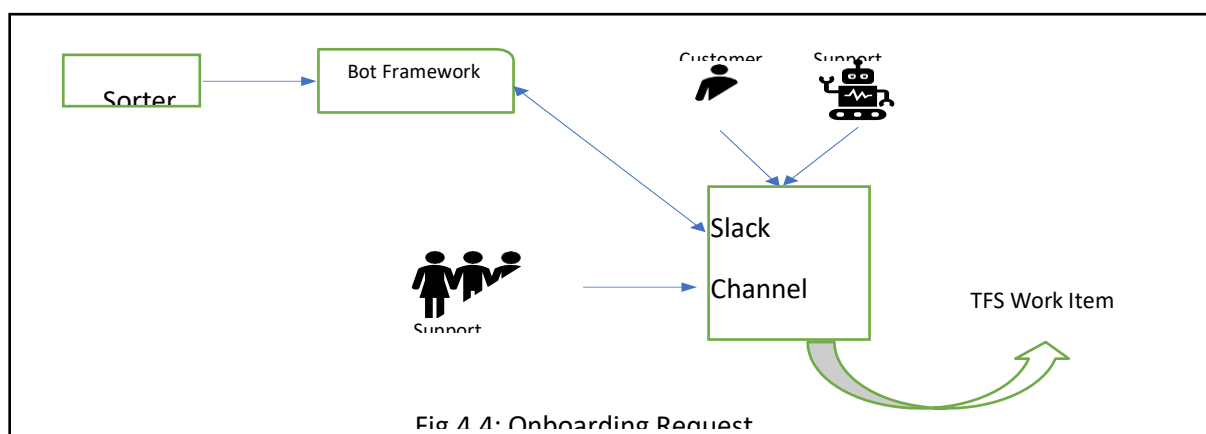
Technology Involved – When the bot picks up the requests it contacts the customer who is seeking assistance for the onboarding of the application onto the Gitlab platform via outlook mailbox. The mail consists of a link to a Slack channel. After the customer joins, the bot is triggered and it asks a set of predefined questions to the user.

After the customer visits the channel and answers the questions, the bot will be triggered again. It will intimate the L1 Support Executive for further assistance. The executive will be sent an email on his/her outlook which would include the link of the Slack channel invite. The executive can join the channel then.

After having a brief idea about the application that needs to be onboarded, the channel can be closed. Now, *Story and Task* creation takes place on the *Team Foundation Server (TFS)*.

Onboarding an application is a lengthy process and so, it is divided into sub-parts which are stored as stories on TFS.

Although in the other branches discussed above, the ticket is closed only after the issue/task/request is resolved, in onboarding requests, ticket is closed even though the issue/request is not yet resolved. Instead, all the information is migrated to TFS and a Feature Id and a Story Id is given to the customer which helps in keeping track of the application progress in terms of percentage onboarded.



Chapter 5 – Proposed Solution

The proposed solution is a bot which has been designed using the .NET framework and has been coded in C# (C Sharp). The bot will use REST API to interact with the SNOW requests and convert them into JSON objects which will need to be handled. This entire chain of events will help the bot is segregating the tickets into different types which can then be handled in their respective manners and the ticket can be closed.

5.1 Working and Pipeline

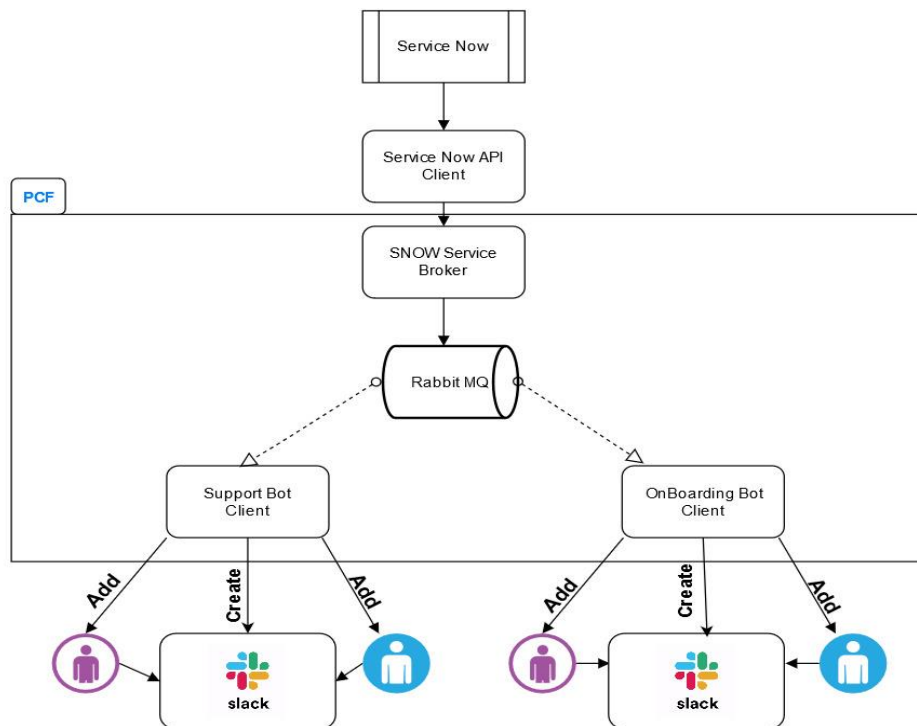


Fig. 5.1: Pipelining of the project

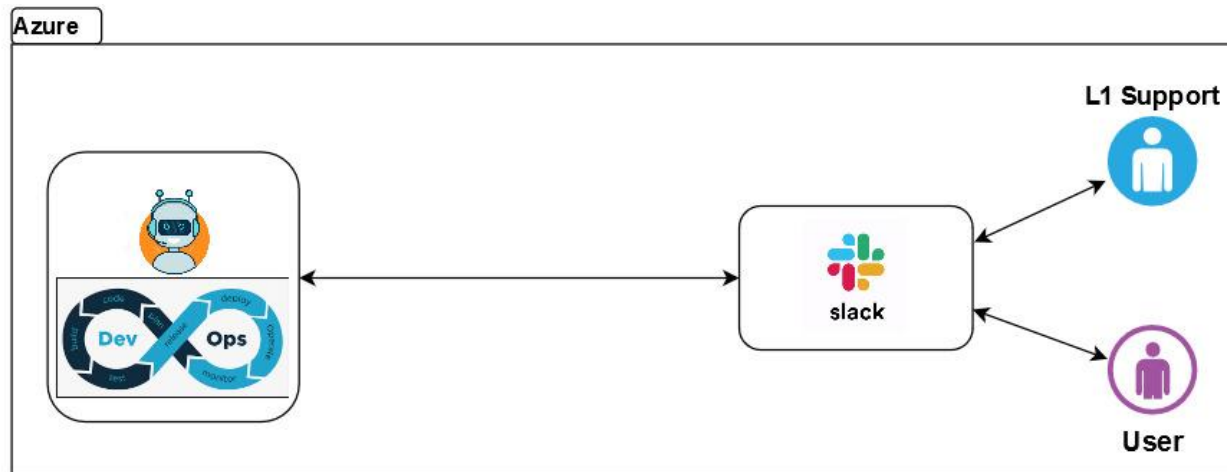


Fig. 5.2: Azure Deployment of Bot

Once the ticket has been raised on the Service Now Page, the Service Now API Client will fetch those tickets and hand them over to the SNOW Service Broker which is responsible for feeding them into a Rabbit Message Queue. The Sorter works on this queue to segregate the tickets and depending upon the type, the ticket is passed onto the Bot client. The client will then create a Slack Channel and add the customer and the L1 support person to it. The Bot client and the Service Broker have been deployed in Pivotal Cloud Foundry (PCF) whereas the created channel has been deployed in Azure.

5.2 The Workflow

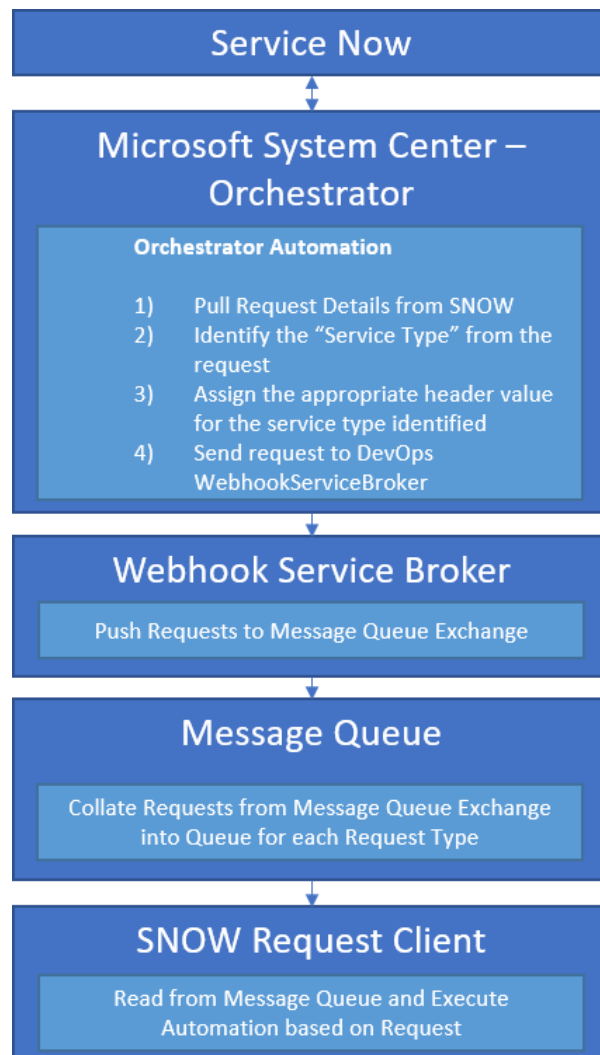


Fig. 5.3: Technologies used in the pipeline

- Service Now Request (Consumer Client)

This client should consume the Service Now request information using API provided by SNOW team and should pass it to the Service Now Broker.

- Service Now Request Update (Status & Comments Client)

This API should be capable of updating Service Now Request's status and its comments.

- Orchestrator API
- Use XML format for sending the Input

- SNOW Service Broker

This broker will consume the SNOW request consumer client and pushes that information to Queue.

- RABBIT MQ

- Exchange Creation
- Queue creation and configuration

- Support Queue Service

This service will consume the Support / On-Boarding Queues, connects to AD service to get DL details and handed over the processing to Support Bot service API (Deployed in Azure).

- Modals
- AD service integration
- Support Bot Service Interaction

- Slack App Channel Creation

Support Bot app should be configured in slack Dell Digital sandbox space.

Key Takeaways:

- Engage SNOW to *Pull, Update and Close* requests
- Read SNOW requests
- Sort the Requests based on Header
- Update SNOW request with TFS story details or comments
- Close SNOW Request
- Slack Channel creation, adding resources and closing channel
- Create BOT to ask questions in the channel and display that information in the channel

- Add initial message and SNOW information in the channel
- Updating channel information in the SNOW and archiving the channel
- Updating channel information in TFS story in case of Onboarding request
- Create TFS user story automatically

5.3 Challenges

Firstly, being a vast project, the work at hand involves a lot of small processes and tasks which need to be synchronized perfectly in order to deliver a bug free product with defined and accurate results.

Each sub task needs to be tested individually for its credibility and if passed has to be merged to the main project and then tested altogether for the compatibility of the function.

As the timeline moved the team incurred many blockheads which hindered the project flow and development. Challenges faced with the development of this project are listed below.

- Dividing up the branches and then those branches into clearly defined tasks which will build up to the major functions of the BOT framework.
- Rendering the SNOW API to fetch tickets. These tickets are highly confidential and the

API needs to be highly secured to pass the security check.

- One of the biggest problems was to create a channel on Slack as it is not a product of Dell and the POC of slack needs to provide the security details so that the key can be set up in the BOT framework to create channels on Slack.
- Another major challenge was to gain access to the Azure license.
- The Team Foundation Server code which was provided to the team was outdated and had to be modified.

Chapter 6 – Tech Stack Overview

6.1 Visual Studio IDE

Visual Studio is an excellent product by Microsoft. It is an Integrated Development Environment (IDE) and is a one stop solution to develop, analyze, debug, test, collaborate and deploy. With the cloud connect and git connect features, we can manage and track changes to the files we are working on.

It provides numerous workloads such as Web & Cloud (ASP .NET, Azure, Python, Node.js), Desktop & Mobile (.NET Desktop, UWP Development, Mobile Development with .NET), Gaming (Unity), etc. The workload which is important to our project is the Bot SDK for .NET.

The *Bot Framework SDK* for .NET is a powerful framework for constructing bots that can handle both free-form interactions and more guided conversations where the user selects from possible values. It is easy to use and leverages C# to provide a familiar way for .NET developers to write bots.

Using the SDK, you can build bots that take advantage of the following SDK features:

- Powerful dialog system with dialogs that are isolated and composable

- Built-in prompts for simple things such as Yes/No, strings, numbers, and enumerations
- Built-in dialogs that utilize powerful AI frameworks such as LUIS.
- FormFlow for automatically generating a bot (from a C# class) that guides the user through the conversation, providing help, navigation, clarification, and confirmation as needed

6.2 Postman

Postman is an API(application programming interface) development tool which helps to build, test and modify APIs. Almost any functionality that could be needed by any developer is encapsulated in this tool. It is used by over 5 million developers every month to make their API development easy and simple. It has the ability to make various types of HTTP requests(GET, POST, PUT, PATCH), saving environments for later use, converting the API to code for various languages(like JavaScript, Python).

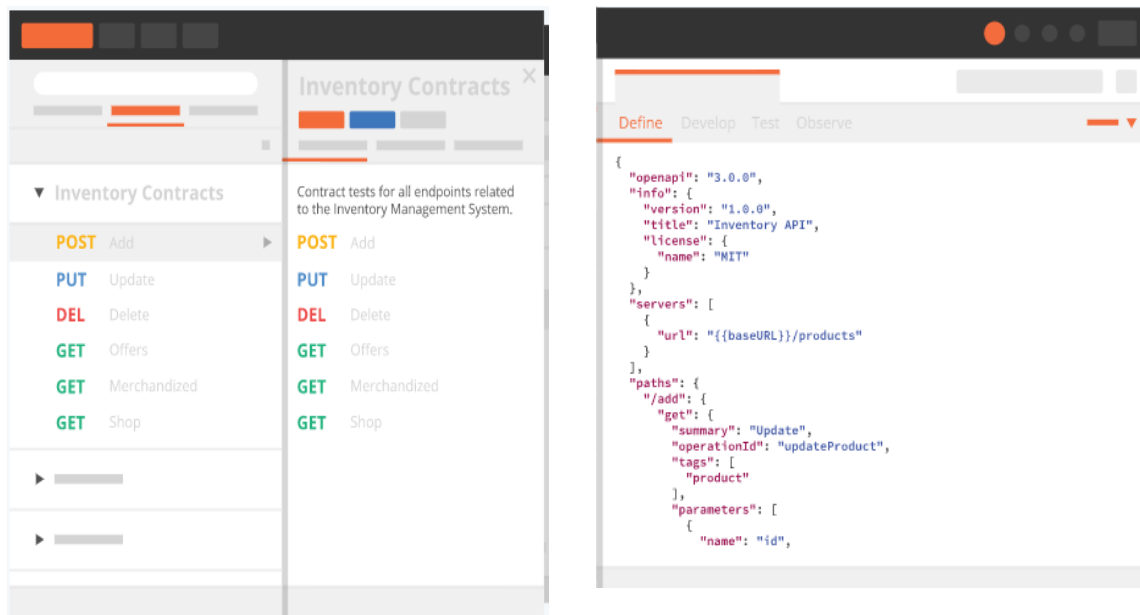


Fig.6.1: Postman functions and the API calling parameters

6.3 Slack

Slack is a popular software to collaborate with team members. Some of the key features are:

- A Channel for every conversation
 - Channels can be divided up by team, project and client or whatever else is relevant to some organization.
 - Team members can join and leave channels as needed, unlike lengthy email chains
 - Threads keep aside conversations from derailing the topic or project at hand.

- The slack app directory has over 2,000 apps which can be integrated to the slack channels for improving productivity
- Slack channels can be deployed with custom bots for catering to the needs of the channel and the organization.
- We can have more custom tools and support via the ease of configurability which Slack provides for the team's unique needs and technology through the Slack API.

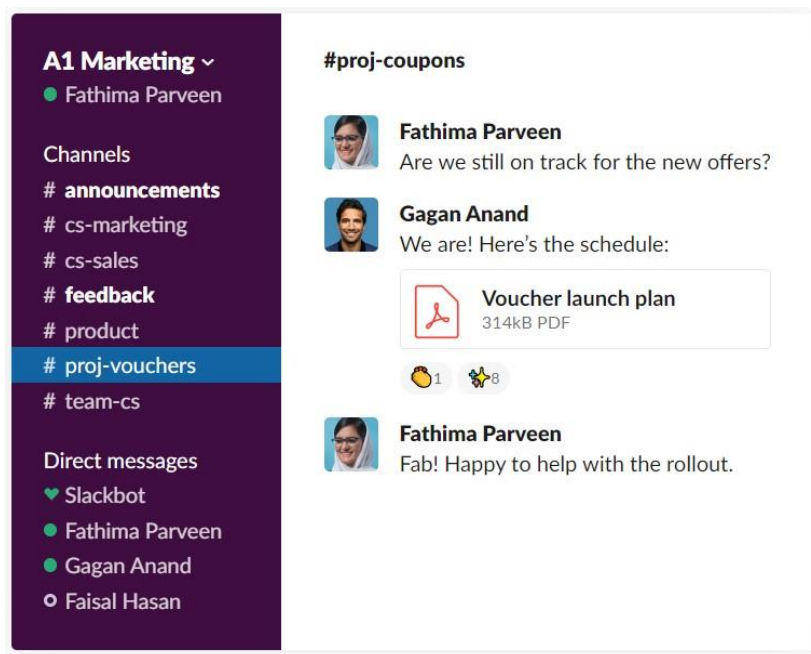


Fig.6.2: Slack



Fig.6.3: Slack Channel invite initiated by the Support Bot

6.4 Team Foundation Server (TFS)

Popularly known as TFS, Team Foundation Server is a product of Microsoft. It is used for source control and helps in managing teams and their code base. TFS offers a combination of version control, issue tracking and application lifecycle management.

Microsoft TFS for source control has been around for more than a decade and has evolved a lot since its inception in 2005. There are professionals in the industry whose entire careers have been dedicated to managing TFS. Such expertise has been necessary because it is quite complicated to administer.

Dell uses TFS for team essential work flows which are managed using stories and tasks. To clearly define the work and maintain a smooth flow, every member is assigned with a story.

As TFS is a source control software, other team members can track commits which have been pushed by his/her team members. However, with the increase in the popularity of Gitlab, TFS has receded to the background.

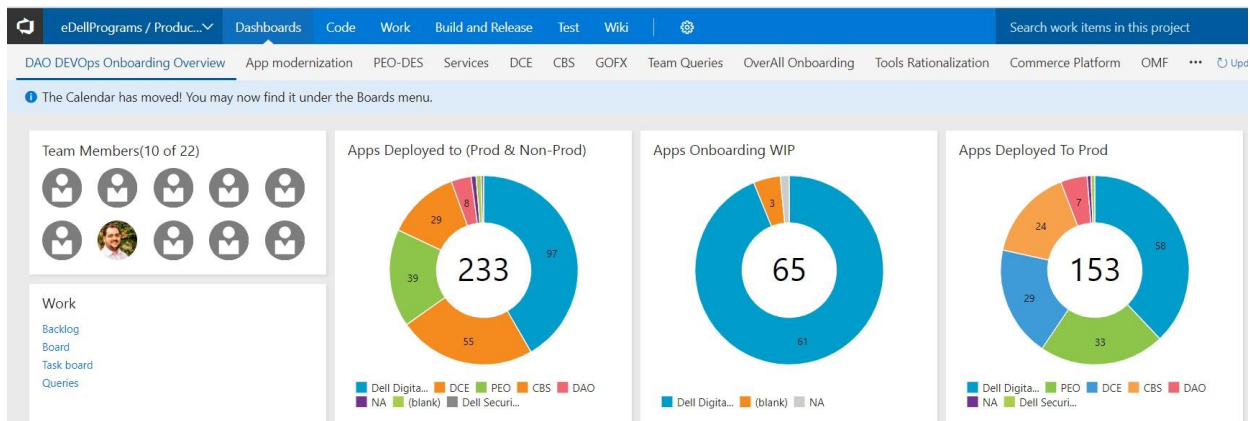


Fig.6.4: Team Foundation Server Dashboard

/eDell/eDellPrograms

Add link

You are adding a link from:

8109864 Onboarding FIL- [Traditi...
Updated 4/28/2020, ● Cancelled

Link type

Affected By

Work item type

Task

Title

OK Cancel

Fig. 6.5: Team Foundation Server Manual Work Item Creation

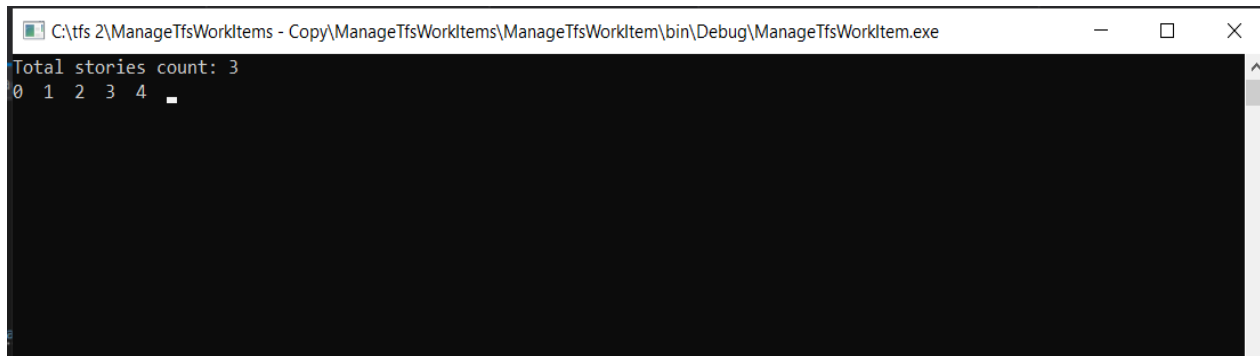


Fig. 6.6: TFS work item creation via a program

6.5 Gitlab

GitLab is a complete open-source DevOps platform, delivered as a single application, fundamentally changing the way development, security and Ops teams collaborate and build software. From idea to production, GitLab helps teams improve cycle time from weeks to minutes, reduce development process costs and decrease time to market while increasing developer productivity.

One of the key features of GitLab is Continuous Integration (CI) & Continuous Delivery (CD). GitLab CI/CD pipelines build, test, deploy, and monitor your code as part of a single, integrated workflow.

Continuous Integration (CI) works to integrate code provided by your team in a shared repository. Developers share the new code in a Merge (Pull) Request. The request triggers a pipeline to build, test, and validate the new code prior to merging the changes within your repository.

The practice of Continuous Delivery (CD) ensures the delivery of CI validated code to your application by means of a structured deployment pipeline.

Together, CI and CD act to accelerate how quickly your team can deliver results for your customers and stakeholders. CI helps you catch and reduce bugs early in the development cycle, and CD moves verified code to your applications faster.

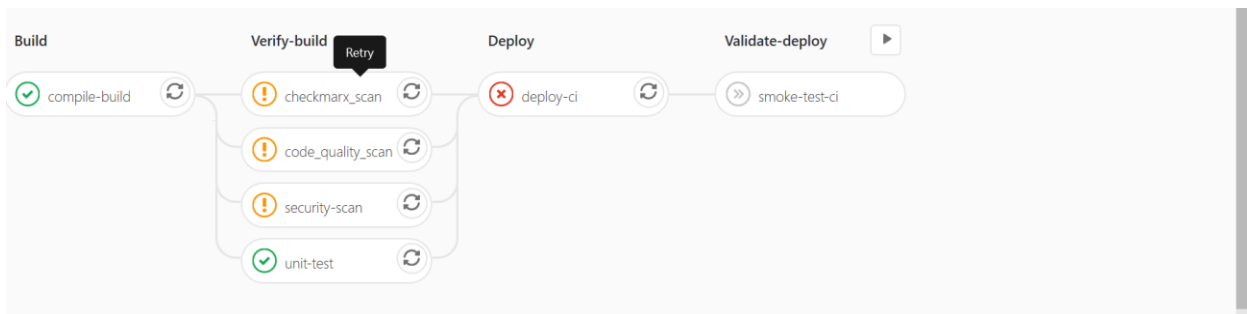


Fig. 6.7: GitLab CI/CD pipeline with jobs and stages

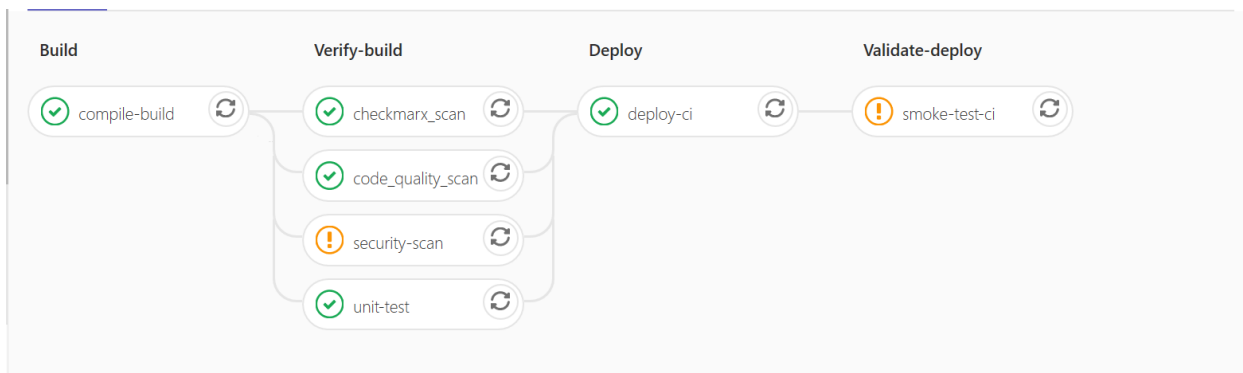


Fig.6.8: GitLab CI/CD pipeline with jobs and stages

6.6 Azure Bot Framework

The main tech stack around which the project is built is *Azure Bot Framework*.

Apart from being reliable, it is also compatible with various pipelining architectures and other softwares with which this project will be interacting.

Azure Bot Service provides an integrated environment that is built for bot development, enabling us to build, connect, test, deploy, and manage intelligent bots, all from one place. Azure Bot Service leverages the Bot Framework SDK with support for C# and JavaScript. Through the use of modular and extensible framework provided by the SDK, tools, templates, and AI services developers can create bots that use speech, understand natural language, handle questions and answers, and more.

Bots provide an unmatched experience that feels less like using a computer that normal people are familiar with and more like dealing with an actual live person - or at least an intelligent robot.

They can be used for repetitive tasks or time taking and less meaningful jobs like gathering profile information and hence, automate systems that may no longer require direct human intervention.

Users converse with a bot using text, interactive cards, and speech.

The flow of developing a bot is simple and can be explained by the following steps.

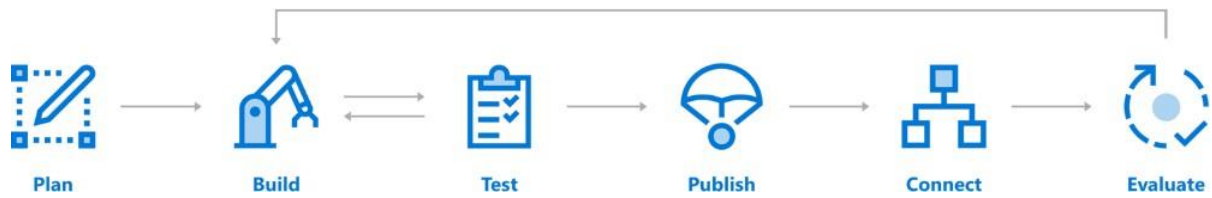


Fig.6.9: Microsoft Bot development stages

6.7 Azure Cloud

Microsoft Azure is a cloud computing service created by Microsoft for building, testing, deploying and managing applications and services through Microsoft-managed data centers. It provides software as a service (SaaS), platform as a service (PaaS) and infrastructure as a service (IaaS) and supports many different programming languages, tools, and frameworks, including both Microsoft-specific and third-party software and systems. Azure cloud has been used extensively in hosting the applications developed during the course of this project.

Chapter 7 – Conclusion

DevOps is helping businesses in a tremendous way. It's bridging the gap between developers' need for change and operations' resist to change and thus creating a smooth path for Continuous Development and Continuous Integration. The project “Support Bot for Automating L1 Triage” was initiated with the sole purpose of reducing the human involvement in the tiring and less innovating work of support assistance for the issues raised. These roles could be changed and the people assigned to it can be put into much more product-oriented roles which will not only benefit them but also benefit the DevOps team and Dell in the long run. This project is aligned towards the goals of achieving the Dell-Digital Way ambition and is a stepping stone towards the vision of Dell Digital.

The project in a nutshell is described as tool or an assistance which will reduce the human involvement into support role by automating some of the tasks. This Support Bot is aimed towards automating tasks like reading the tickets incoming and then sorting them into their respective branches as per the header reads. Post sorting a queue is maintained in each of these branches for the incoming messages. These messages or tickets are then acted up with the respective flow and guideline as provided to the BOT.

If the BOT works as intended in the preliminary stages then it can be generalized and put to work into other fields where the automation is required and this would not only help the team to upscale the productivity but also make the team members focus on work

which is more crucial.

The BOT will set new standards of dealing with the tickets that are raised on a daily basis.

The application is built with an idea and scope for future integration and development and also keeping in mind the idea of reusability of the services like the API request to create channel on demand, Then comes the integration of API for automatic story creation which can be used in other applications because story and task creation on TFS is used profusely.

The resources used in the development of this project have a very high value for future implementation.

Chapter 8 – Future Scope

8.1 Migration to Microsoft Teams

The project is an effort to incorporate the idea of automation. This project can be thought of as the first step of this journey of automation wherein the basic idea has been to remove human intervention in the possible areas.

During the course of this endeavour, we figured that the work and complexity of some technologies is conducive towards automation and if automated, they would definitely improve the productivity of the team.

Since this project is just the beginning to the automated future, there are absolutely infinite possibilities of improvements and addition to it.

The next major step which we are leaving out in the initial phases of the development is the migration of channel creation from Slack to *Microsoft Teams* and for this we would have to work with another techstack called *Microsoft Graph API*. This migration will prove more logical in the later stages as currently DevOps team is communicating over Slack but as the company is gradually moving towards Microsoft Teams, it would be the new standard.

8.2 Addition of Non-Compatible Tools

Some tools supported by the DevOps team are currently non-compatible with the bot. The next step will be automation at a hundred percent mark where all the tools available in each of the categories will be linked to the Support.

The vision of this Support Bot is lot more than automation of support tools and backend plugins. It is initiated with an ambition of completely removing human intervention in L1 Support Triage.

8.3 The Future of ChatBots

The natural conclusion of evolution of ChatBots will be the requirement to update the bots used in this project. This will in turn help in unlocking greater potential for automation. Hence, this project will remain in continuous developmental stage.

Chatbots, like other AI tools, will be used to further enhance human capabilities and free humans to be more creative and innovative, spending more of their time on strategic rather than tactical activities.

In the near future, when AI is combined with the development of 5G technology, businesses, employees, and consumers are likely to enjoy enhanced chatbot features such as faster recommendations and predictions, and easy access to high-definition video conferencing from

within a conversation. These and other possibilities are in the investigative stages and will evolve quickly as internet connectivity, AI, NLP, and ML advance. Eventually, every person can have a fully functional personal assistant right in their pocket, making our world a more efficient and connected place to live and work.

Chapter 9 – References

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